



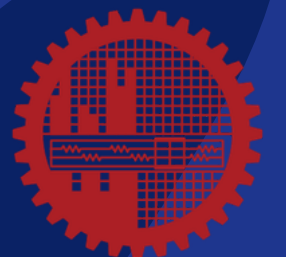
ATLAS

✦ ADAPTIVE TUTORING VIA LEARNING AND SEMANTICS ✦

“Democratizing Personal Tutoring for the Next Billion Students”



FINAL PRESENTATION



MEET SABBIR



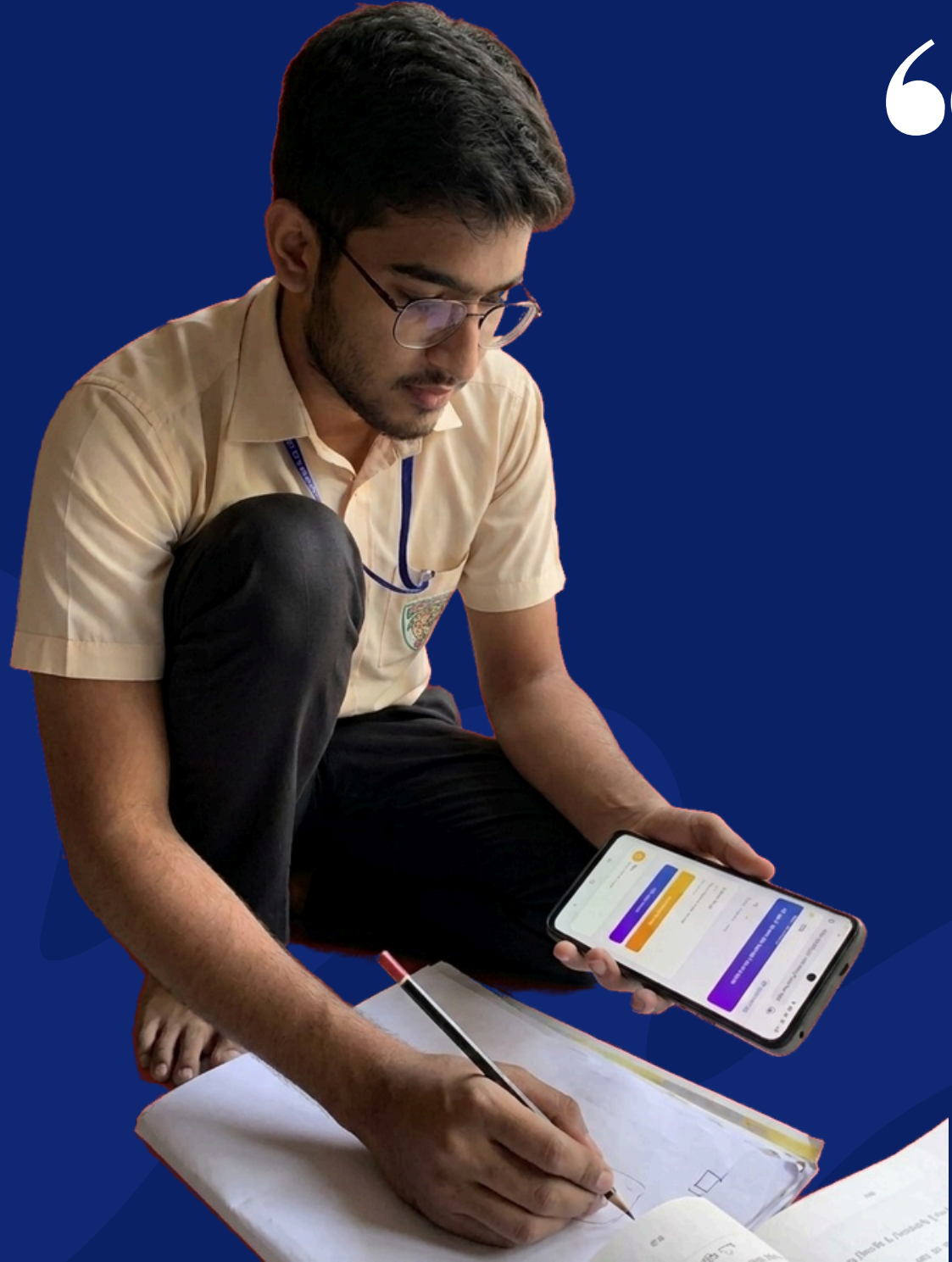
“

আঃ আব্বির ইমদাদ
বয়স = 3
স্কুলের নাম = প্রাথমিক প্রাথমিক প্রাথমিক
বিদ্যালয়
বছরঃ ২০২০

”

- loves math but feels helpless at times
- stuck on a simplification problem
- not "bad at math"
- specifically didn't understand how fractions interacted with whole numbers

MEET SHAKSHOR



“ক্যালকুলাসের মূল জিনিসগুলো মোটামুটি ভালোই বুঝি, ইন্টিগ্রেশনও করতে পারি। কিন্তু অংকে সাইন-কস বা বিপরীত ত্রিকোণমিতি এলেই কেমন যেন আটকে যাই। আর এত বড় কোচিং ব্যাচে স্যারকে যে আলাদা করে নিজের এই দুর্বলতাটা বোঝাব, সেই সুযোগও নেই।”

”

PROBLEM IDENTIFICATION

PROBLEM ANALYSIS



RESULTS

- **"SWISS CHEESE" LEARNING:** STUDENTS ACCUMULATE SMALL HOLES IN THEIR KNOWLEDGE (LIKE SABBIR'S FRACTION GAP) THAT EVENTUALLY CAUSE THE WHOLE STRUCTURE TO COLLAPSE IN HIGHER GRADES.
- **ROTE MEMORIZATION:** WITHOUT UNDERSTANDING, STUDENTS MEMORIZE ANSWERS JUST TO PASS EXAMS.

CORE PROBLEM

THE "2 SIGMA PROBLEM" (BLOOM, 1984): RESEARCH PROVES THAT STUDENTS WITH A PERSONAL TUTOR PERFORM 2 STANDARD DEVIATIONS (98%) BETTER THAN THOSE IN A STANDARD CLASSROOM.



ROOT CAUSES



THE ECONOMIC BARRIER



THE STRUCTURAL BARRIER



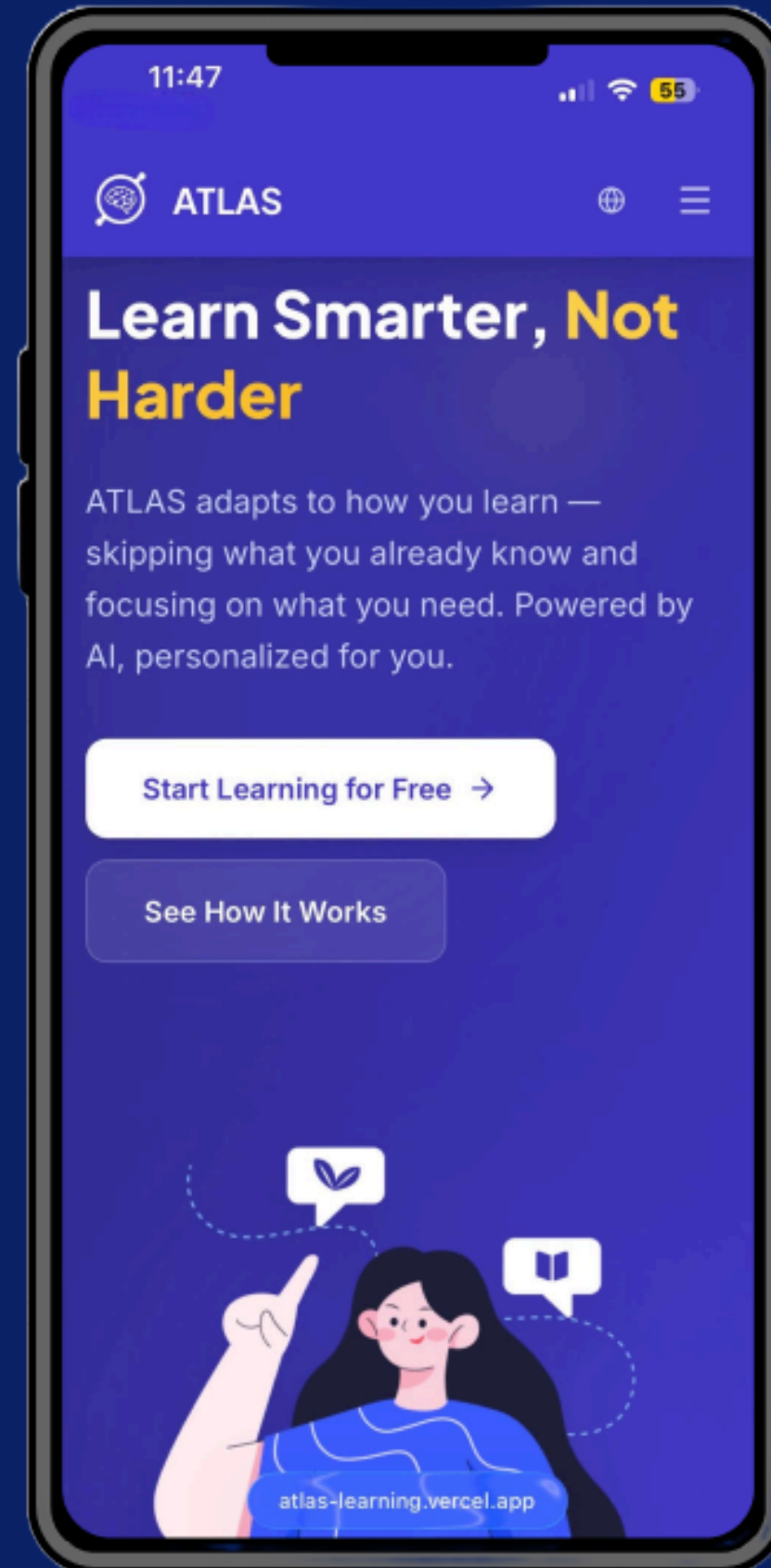
THE PEDAGOGICAL BARRIER

THE “BIG IDEA,”



IT DIAGNOSES

LIKE A DOCTOR, IT FINDS THE EXACT
ROOT CAUSE OF YOUR MISTAKES
(E.G., "YOUR SKILL IN ALGEBRA IS
ONLY 20%!")



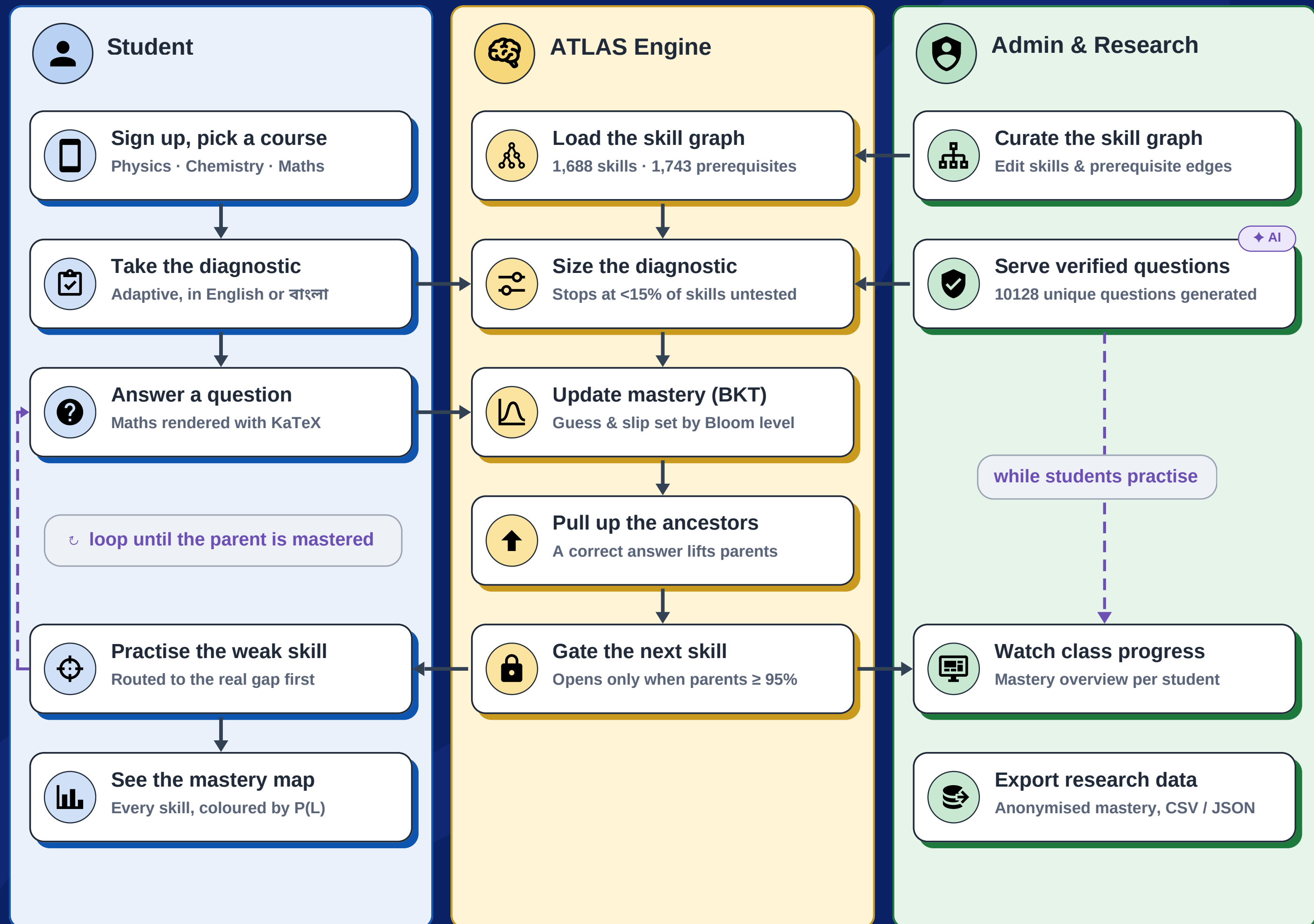
IT ACTS

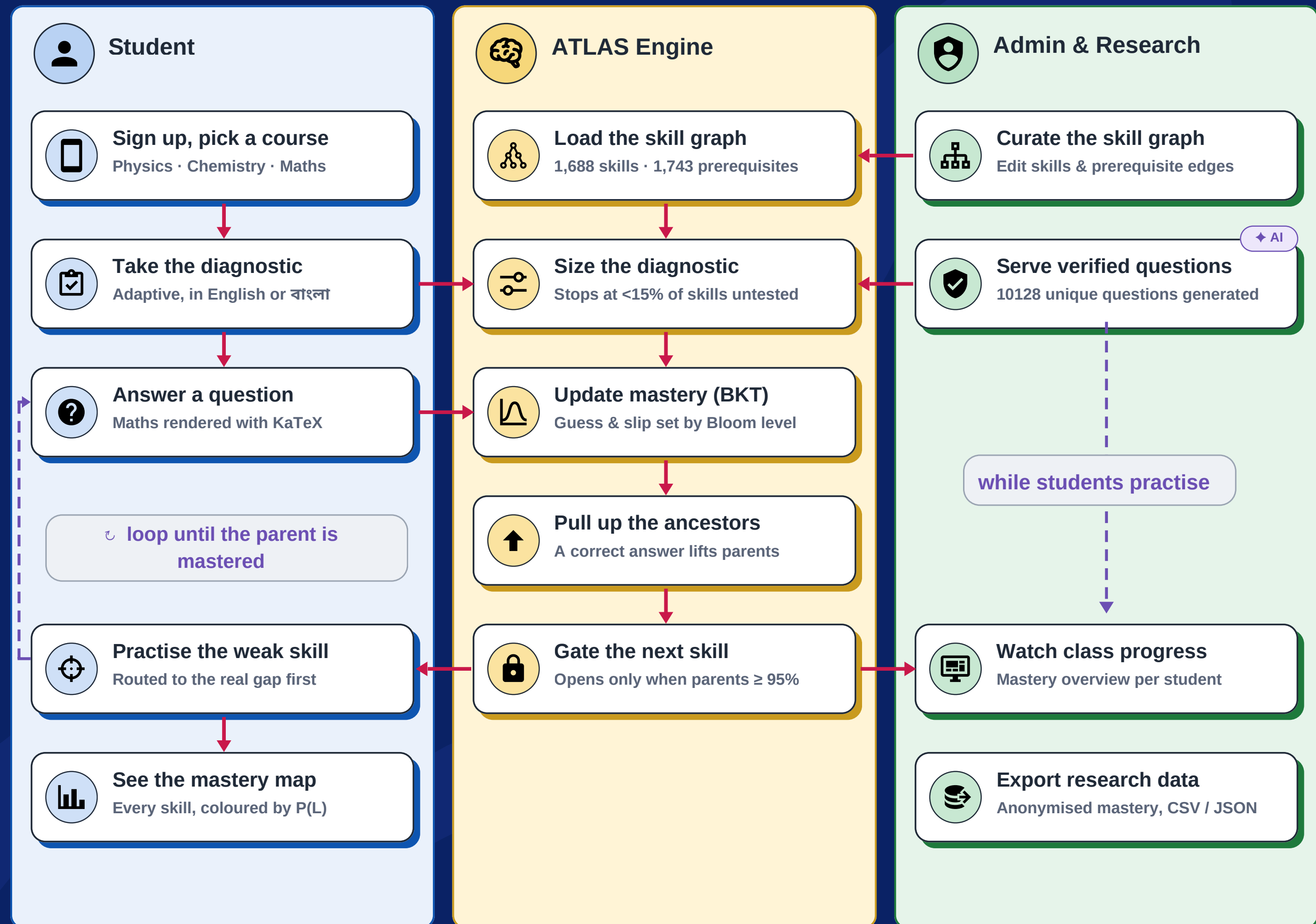
IT DOESN'T JUST SAY "WRONG." IT
HELPS RECTIFY THE MISTAKE BY
ADDRESSING THE ISSUE IN REAL TIME

IT UNDERSTANDS

IT CHANGES THE DIFFICULTY AND
TYPE OF QUESTION IN REAL-TIME
BASED ON THE STUDENT'S STRUGGLE

USER JOURNEY





TIME FOR THE DEMO

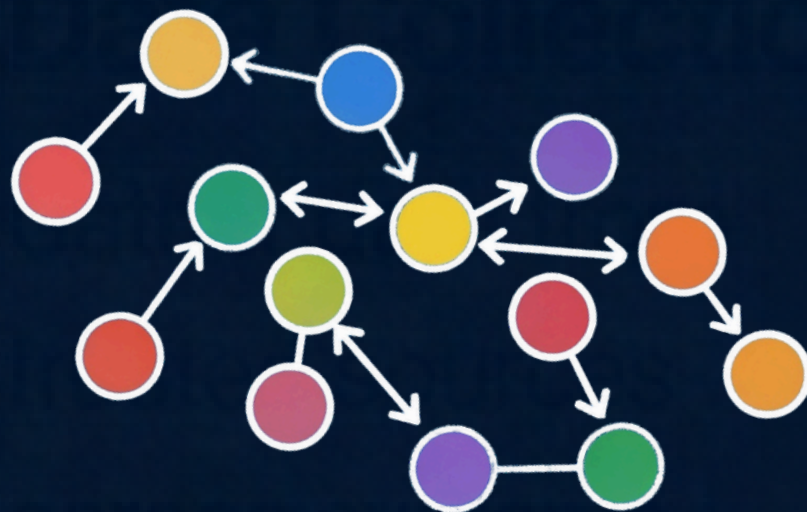


UNDER THE HOOD

FULL PIPELINE

01

Prerequisite Skill Graph



02

Question Bank Generation

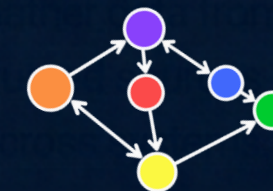


03

Adaptive Learning Engine

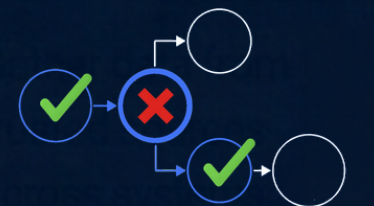
03a

Ancestor Pullup and
Conjunctive Gating



03b

Prerequisite Spillover



1688
SKILLS

183
TOPICS

1743
PREREQ EDGES

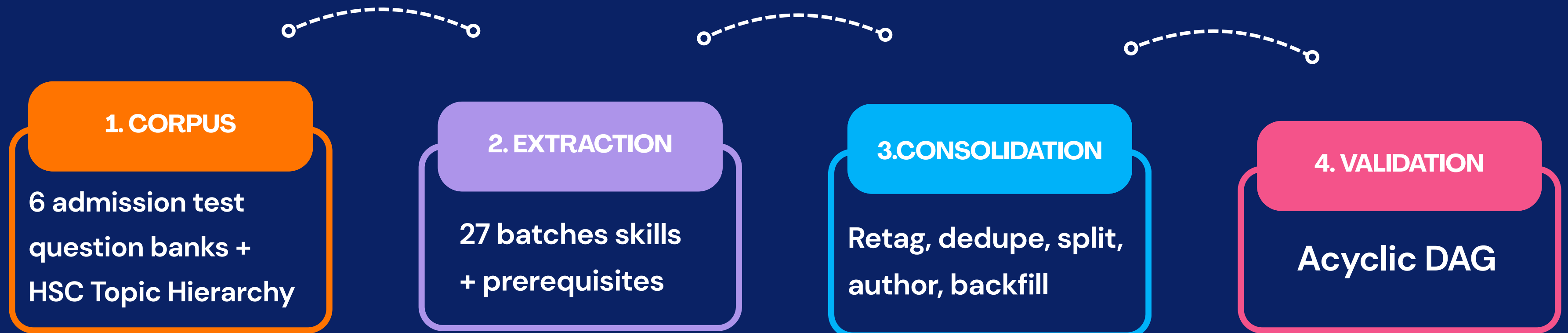
0%
UNCONNECTED

87
FRAGMENTS

PREREQUISITE SKILL GRAPH



HOW WE BUILT IT

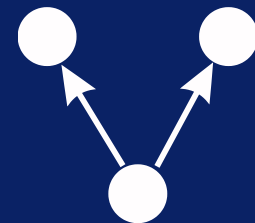


THE RESULT



SKILLS

1688



PREREQUISITE EDGES

1743



SYLLABUS TOPICS
POPULATED

183

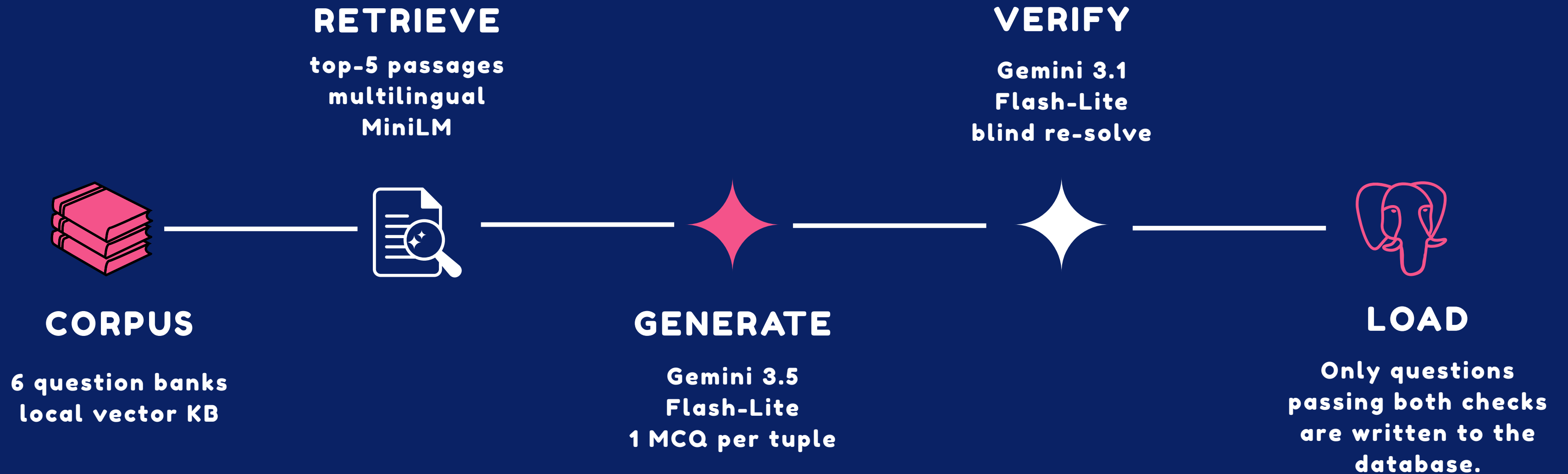


MAX DAG DEPTH

9

QUESTION BANK GENERATION

THE PIPELINE



PROMPT DESIGN

1. QUESTION GENERATION

prompt

You are an expert MCQ writer for Bangladesh admission exams. Generate NEW questions for the requested skill from the retrieved references – do NOT copy them verbatim. Output a JSON array: 4 options A–D, one correct, explanation for each.

✦ **gemini-3.5-flash-lite**

- Role prompting
- retrieval grounding
- Explicit anti-copy
- strict JSON schema
- Bloom-level conditioning
- Bangla + LaTeX rules
- Distractors

2. INDEPENDENT VALIDATION

prompt

You are an expert MCQ writer for Bangladesh admission exams. Generate NEW questions for the requested skill from the retrieved references – do NOT copy them verbatim. Output a JSON array: 4 options A–D, one correct, explanation for each.

✦ **gemini-3.1-flash-lite**

- Independent second pass
- exact-skill relevance
- Blind re-solve
- marked answer not trusted
- Separate model from the generator
- Structured verdict: `on_topic` + `correct_answer`

SCALE

1,688 skills × 6 Bloom levels = 10,128 tuples

Every skill is generated at all six levels, because the engine always asks one level above the learner's current band.

ADAPTIVE LEARNING ENGINE: BAYESIAN KNOWLEDGE TRACING

ONE NUMBER PER SKILL

EVERY SKILL CARRIES A PROBABILITY THE LEARNER HAS MASTERED IT, UPDATED ON EVERY SINGLE ANSWER.

$P(L_0)$

prior mastery

$P(T)$

learning rate

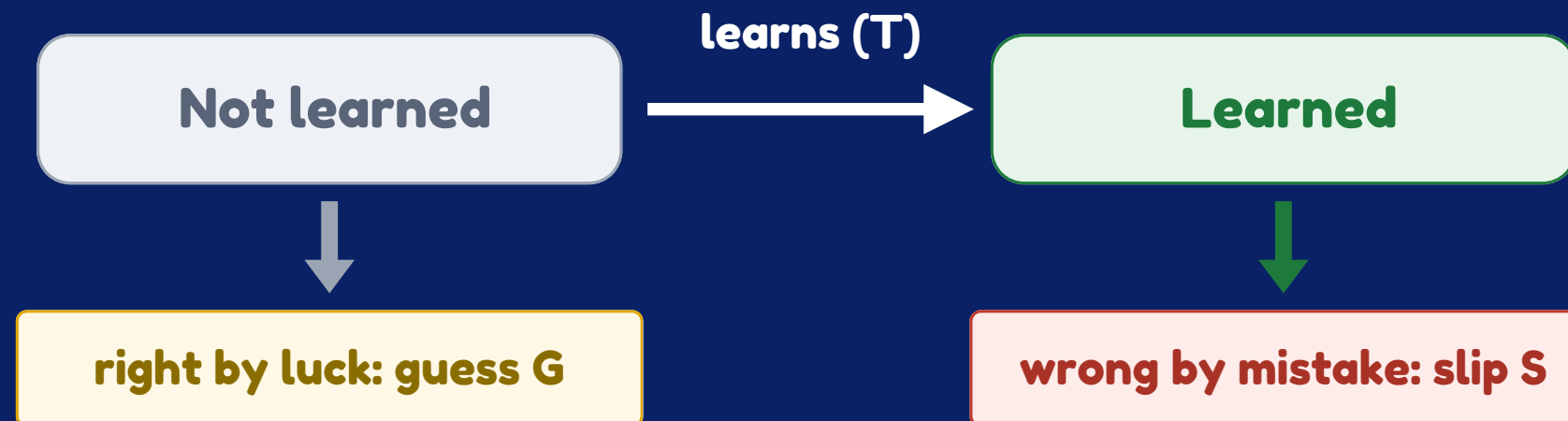
$P(S)$

slip – knows it, answers
wrong

$P(G)$

guess – doesn't know,
answers right

ATLAS can't see whether a student knows a skill. It only sees their answers, so it keeps a probability $P(L)$ for every skill and updates it after each one.



after a correct answer	after a wrong answer
$P(L \checkmark) = \frac{P(L)(1 - S)}{P(L)(1 - S) + (1 - P(L))G}$	$P(L \times) = \frac{P(L) \cdot S}{P(L)S + (1 - P(L))(1 - G)}$

then learning: $P(L_{n+1}) = P(L | \text{obs}) + (1 - P(L | \text{obs})) \cdot T$



Prerequisite gate

$T = 0.10$ only once every parent reaches 0.95



Ancestor pull-up

Correct answers lift parents to the child's $P(L)$

EVERY ANSWER, THREE PHASES

BKT UPDATE

Mastery of the answered skill moves, using that question's guess and slip.



ANCESTOR PULL-UP

A correct answer raises every ancestor to $\max(\text{parent}, \text{child})$. Wrong answers never push down.



CONJUNCTIVE GATE

slip – knows it, answers wrong

PREREQUISITE SPILLOVER

RULE A

3 of the last 5 answers were wrong on the same missing prerequisite.

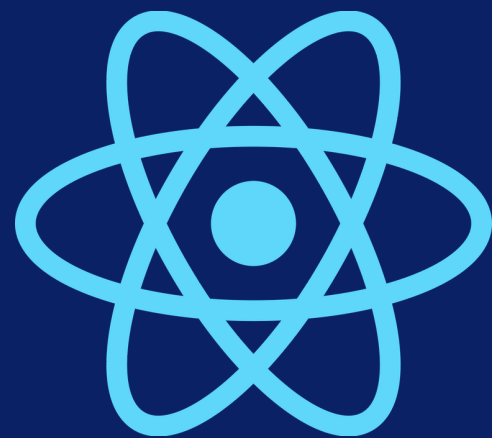
RULE B

Accuracy below 40% over the last 8, and a candidate prerequisite below 0.60.

EITHER FIRES → inject 2 questions on that prerequisite, then resume the topic.

OVERALL TECHNOLOGY STACK

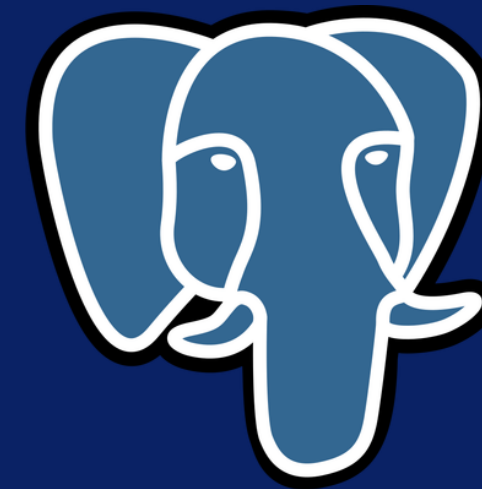
FRONTEND



API GATEWAY



DATABASE



INFERENCE ENGINE



MEET TEAM **ATLAS**

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AURCHI CHOWDHURY



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